

RFM Customer Segmentation

Dunnhumby — The Complete Journey

2,500 Households · 711 Days · 2.6M Transactions

RFM

45.4%

Champions' revenue share

42.6%

Churn rate

\$8.06M

Total 2-year revenue

The Business Problem

A retail grocer treating 2,500 households identically — same campaigns, same coupons, no differentiation

01 Who are our most valuable customers?

And how do we make sure we never lose them?

02 Who is quietly drifting away?

42.6% of households show measurable churn signals.

03 Are we wasting promotion spend?

Are campaigns reaching customers who'd buy anyway?

Dataset & Methodology

Dataset

- ▶ **transaction_data.csv**
2.6M rows — core purchase data
- ▶ **hh_demographic.csv**
801 rows — 32% household coverage
- ▶ **campaign_table.csv**
7,208 rows — campaign targeting
- ▶ **campaign_desc.csv**
30 rows — campaign windows
- ▶ **coupon_redempt.csv**
2,318 rows — redemption events
- ▶ **product.csv**
92K rows — product hierarchy

RFM Scoring — Quintile Binning

R

Recency

Days since last purchase
Score 5 = shopped recently
Score 1 = 658 days absent
Ref day = 712, labels inverted

F

Frequency

Distinct shopping trips
Score 5 = 1,300 trips
Score 1 = 1 trip
rank(method='first') for ties

M

Monetary

Total spend over 2 years
Score 5 = \$38,319
Score 1 = \$8
Excludes \$0 promo items

RFM Score = R + F + M · Range: 3 (worst) → 15 (best) · `pd.qcut()` quintile binning

8 Customer Segments

Rule-based classification using np.select() — 0 uncategorised households

Segment	Households	Avg RFM Score	Revenue %	Rule
Champions	509	13.84	45.4%	$R \geq 4, F \geq 4, M \geq 4$
Loyal	330	11.53	17.0%	$F \geq 4, R \geq 3$
At Risk	565	8.36	18.4%	$R \in [2,3], F \geq 2, M \geq 2$
Potential Loyalists	306	9.81	8.5%	$R \geq 4, F \in [2,3]$
Lost	500	4.73	8.1%	$R = 1$
One-time Buyers	198	5.07	1.9%	$R \in [2-4], F = 1$
Low Value	62	5.45	0.4%	$R \in [2,3], F \geq 2, M = 1$
New Customers	30	7.33	0.2%	$R = 5, F = 1$

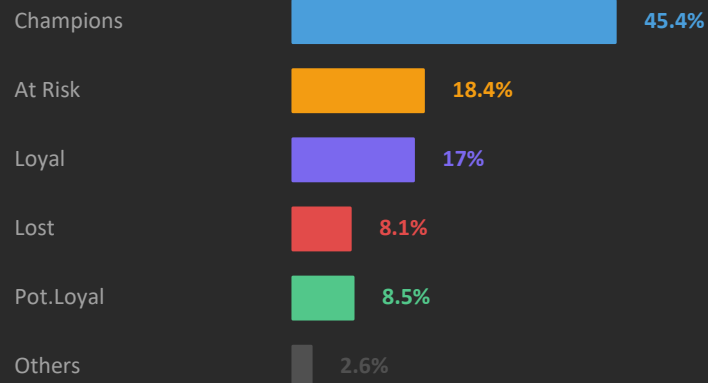
Champions dominate revenue

45.4%

of total revenue from Champions

who are just 20.4% of households

Revenue by Segment



💡 Champions visit every 2.1 days on average — 246 trips over 2 years at \$29.22/trip. Frequency is their superpower, not basket size. One-time Buyers spend \$36.56/trip but made that trip exactly once.

Key Finding 02

42.6% of households show churn signals

At Risk

565 households · 18.4% of revenue

Still active but visit frequency declining. Highest recovery ROI.

Avg Recency

13.1 days

Avg R/F/M

2.39 / 2.88 / 3.09

Lost

500 households · 8.1% of revenue

104 days since last visit. Brand preference unchanged — left for experiential reasons.

Avg Recency

104.2 days

Avg R/F/M

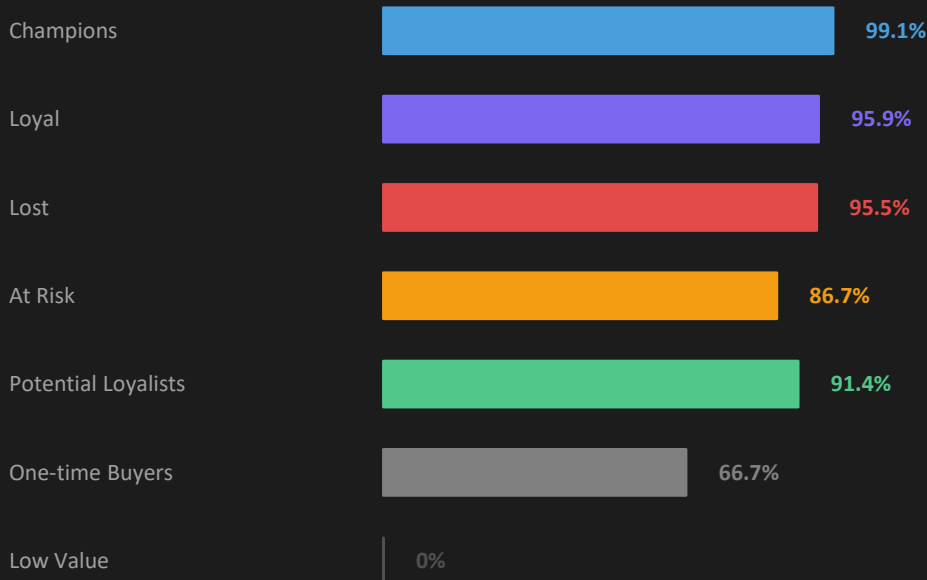
1.00 / 1.83 / 1.90

At Risk + Lost = 42.6% of households · At Risk: 18.4% of revenue · Lost: 8.1% of revenue

Key Finding 03

916 households never received a single campaign

Campaign Targeting Rate by Segment



⚠ Response Rate Caveat

Overall campaign response = 97.9%

This is NOT evidence of effectiveness.

Champions visit every 2.1 days on average (Page 2 card). The average campaign window is 47 days. Champions shop continuously regardless of whether they received a campaign.

97.9% measures shopping frequency — not campaign lift.

TypeB campaigns (63% of all) reach only 40.9% of households · TypeC reaches just 15.9%

Key Finding 04

Coupons reward loyalty that already exists

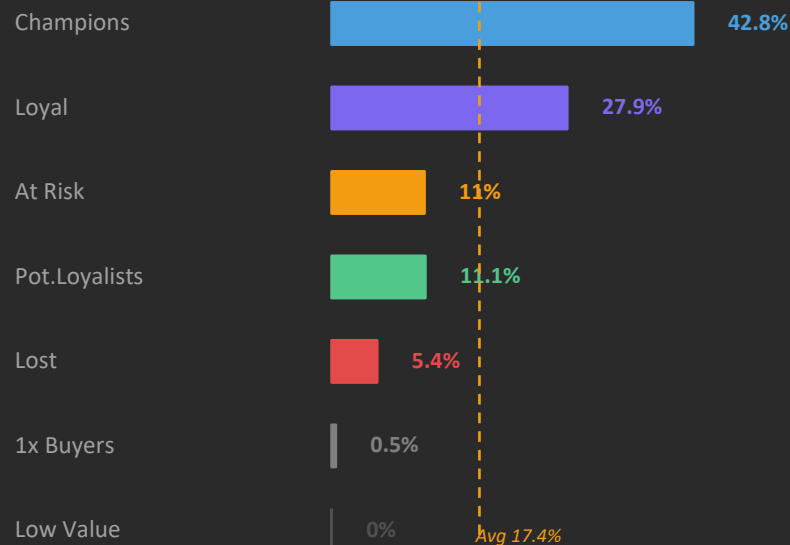
60.1%

of all coupon redemptions

absorbed by Champions

who are just 20.4% of households

Coupon Redemption Rate by Segment



Send a coupon to a Lost customer → 1 in 20 will use it. Send one to a Champion → they'd buy anyway. Coupons are changing zero behaviour.

Data Limitations

Responsible analysis acknowledges what the data cannot tell us

01 Demographic coverage — 32% only

Only 801 of 2,500 households have demographic data. Income band and family size findings are indicative — not universal. Any marketing decision needs validation against a fuller source.

02 Assumed calendar dates

The dataset uses integer day numbers (1–711). I assumed Day 1 = January 1st 2017. If incorrect, calendar labels shift — but all RFM calculations are unaffected.

03 Static segmentation

RFM scores reflect the full 2-year window. A customer active in Year 1 who churned in Year 2 may be scored differently than their recent behaviour suggests. Rolling RFM would be more accurate.

04 Frequency outlier

One household made 1,300 trips over 711 days — 1.8 trips/day. Likely a commercial account or data artefact. Retained in analysis but flagged.

05 No campaign control group

Every targeted household received a campaign. Without a control group we cannot measure what would have happened without it. Response rates cannot be interpreted as causal lift.

4 Actionable Recommendations

01

Reduce Champion campaign spend by 30–40%

Champions visit every 2.1 days (Page 2 card) and buy without incentives. Redirect budget to loyalty recognition — early access, personalised rewards. Higher emotional ROI at lower cost.

→ 99.1% of Champions already targeted — room to reallocate

02

Urgently close the At Risk targeting gap

565 households, 18.4% revenue, 13.25% never received a campaign (Page 3 dashboard). Still active, still spending — highest recovery ROI. Personalise around their top department from Page 2 treemap.

→ At Risk is the largest segment — 565 households drifting

03

Redesign coupon strategy around Potential Loyalists

Champions absorb 60.1% of coupon value (Page 3) but buy anyway. Redirect to Potential Loyalists (306 households, recency 2.4 days from Page 2). The right nudge converts occasional to regular.

→ Build loyalty that doesn't yet exist

04

Expand TypeC and TypeA reach, reduce TypeB concentration

TypeB (63% of campaigns) has only 40.9% household coverage. TypeC reaches just 15.9% of households. Both verified from Page 3 campaign type filter. Expand reach to 916 never-targeted households.

→ Broader funnel without increasing total spend

The data tells a consistent story.

Over-investing in customers who are already loyal.
Under-investing in customers who are at risk or underdeveloped.

45.4%

Champions' revenue share

42.6%

Households showing churn signals

916

Households never targeted

60.1%

Coupon value to Champions

The RFM framework provides the intelligence to change that allocation.